Linear regration

1. What is Linear Regression?

Linear Regression finds the **best-fit straight line** that represents the relationship between independent variables (features) and the dependent variable (target).

Example:

Years of Experience	Salary
1	30000
2	35000
3	42000
4	50000
5	56000

As experience increases, salary generally increases. Linear regression learns this relationship and predicts salaries for new experience values.

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2. Types of Linear Regression

A. Simple Linear Regression

One input feature.

Example

Experience → Salary

Equation:

y=mx+by = mx + by=mx+b
m (slope)
1.0
m (slope)
b (y-intercept)
5.
b (y-intercept)
Y=β0+β1XY=\beta_0+\beta_1XY=β0+β1X

Where

- **Y** = Predicted value
- **X** = Independent variable
- **β_0** = Intercept
- **β_1** = Slope (Coefficient)

Example

Salary=25000+6000×ExperienceSalary=25000+6000\times ExperienceSalary=25000+6000×Experience

If Experience = 5

Salary = 25000 + 6000 × 5

= ₹55,000

B. Multiple Linear Regression

Uses multiple features.

Equation

Y=β0+β1X1+β2X2+⋯+βnXnY=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_nY=β0+β1X1+β2X2+⋯+βnXn

Example

Predict House Price using

- Area
- Bedrooms
- Bathrooms
- Age of House

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3. Goal of Linear Regression

The goal is to find the **best-fit line** that minimizes prediction errors.

The algorithm tries to reduce the distance between

Actual Value

and

Predicted Value

This distance is called **Error** or **Residual**.

Residual:

Residual=Actual−PredictedResidual = Actual - PredictedResidual=Actual−Predicted

---

4. What is Best Fit Line?

Suppose actual points are

```
*
      *
           *
                *
```

Linear Regression draws

```
----------------------
```

The line should pass as close as possible to all points.

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5. Cost Function

Linear Regression minimizes the Cost Function.

The most common cost function is the **Mean Squared Error (MSE)**.

Formula:

MSE=1n∑i=1n(yi−y^i)2MSE=\frac1n\sum_{i=1}^{n}(y_i-\hat y_i)^2MSE=n1i=1∑n(yi−y^i)2

where

- yiy_iyi = Actual value
- y^i\hat y_iy^i = Predicted value
- n = Number of observations

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6. Assumptions of Linear Regression

1. Linear relationship
2. No multicollinearity
3. Homoscedasticity (constant variance)
4. Normal distribution of residuals
5. Independence of observations
6. No significant outliers

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7. Model Training Process

```
Dataset
      ↓
Train-Test Split
      ↓
Train Linear Regression
      ↓
Predict Test Data
      ↓
Evaluate
```

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8. Practical Example

Suppose

Experience	Salary
1	30000
2	35000
3	40000
4	47000
5	52000

Train Model

↓

Model learns

```
Salary = 25000 + 5500 × Experience
```

Now

Experience = 6

Prediction

```
Salary = 25000 + 5500×6

=58000
```

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9. Performance Metrics

A. MAE (Mean Absolute Error)

Formula

MAE=1n∑i=1n|yi−y^i|MAE=\frac1n\sum_{i=1}^{n}|y_i-\hat y_i|MAE=n1i=1∑n|yi−y^i|

What it does

- Calculates the average absolute difference between actual and predicted values.
- Does **not** square errors.
- Every error has equal importance.

Example

Actual

```
100
120
150
```

Predicted

```
90
125
140
```

Errors

```
10
5
10
```

MAE

```
=(10+5+10)/3

=8.33
```

Interpretation

The model is wrong by **8.33 units on average**.

Advantages

- Easy to understand
- Not highly affected by outliers

Disadvantages

- Large and small errors receive similar importance

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10. MSE (Mean Squared Error)

Formula

MSE=1n∑i=1n(yi−y^i)2MSE=\frac1n\sum_{i=1}^{n}(y_i-\hat y_i)^2MSE=n1i=1∑n(yi−y^i)2

Example

Errors

```
10
5
10
```

Squared Errors

```
100
25
100
```

Average

```
225/3

=75
```

Interpretation

The average squared error is **75**.

Advantages

- Penalizes large mistakes more heavily
- Smooth function for optimization

Disadvantages

- Units are squared, making interpretation less intuitive

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11. RMSE (Root Mean Squared Error)

Formula

RMSE=MSERMSE=\sqrt{MSE}RMSE=MSE

Example

`MSE=75`

`RMSE=√75`

`=8.66`

Advantages

- Same unit as the target variable
- Easier to interpret than MSE

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12. R^2 Score (Coefficient of Determination)

Formula

R2=1−∑(yi−y^i)2∑(yi−y¯)2R^2=1-\frac{\sum(y_i-\hat y_i)^2}{\sum(y_i-\bar y)^2}R2=1−∑(yi−y¯)2∑(yi−y^i)2

Meaning

Measures how much of the variation in the target variable is explained by the model.

Interpretation

R^2 Score	Meaning
1.00	Perfect prediction
0.95	Excellent
0.90	Very Good
0.80	Good
0.60	Average
0.00	Model explains nothing
Negative	Model is worse than predicting the mean

Example

```
R² = 0.92
```

The model explains **92% of the variation** in the target variable.

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13. Comparison of Metrics

Metric	Formula	Lower/Upper Better	Sensitive to Outliers	Unit
MAE	Average of absolute errors	Lower	Low	Same as target
MSE	Average of squared errors	Lower	High	Squared unit
RMSE	Square root of MSE	Lower	High	Same as target
R^2	Explained variance	Higher	No (indirectly affected through residuals)	No unit

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14. Python Practical

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
```

```
)
import numpy as np

# Load Dataset
df = pd.read_csv("Salary_Data.csv")

# Features and Target
X = df[['YearsExperience']]
y = df['Salary']

# Split Data
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Train Model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluation
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("Intercept:", model.intercept_)
print("Coefficient:", model.coef_[0])

print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R2 :", r2)
```

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15. Real-World Applications

- House price prediction
- Salary prediction
- Sales forecasting
- GDP forecasting
- Stock trend analysis (for simple trend modeling)
- Electricity demand prediction
- Insurance premium estimation
- Medical cost prediction
- Student marks prediction
- Business revenue forecasting

Interview Questions

1. What is Linear Regression?
2. What is the difference between Simple and Multiple Linear Regression?
3. What is the role of the intercept and coefficient?
4. What is overfitting and underfitting?
5. Explain MAE, MSE, RMSE, and R^2 Score.
6. Why is MSE preferred as a loss function during training?
7. What are the assumptions of Linear Regression?
8. How do you detect multicollinearity?
9. When would you choose MAE over RMSE?
10. How do you interpret an R^2 score of 0.85?